

Inventory Financing Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Financing Burden (IFB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IFB achieves an annualized gross (net) Sharpe ratio of 0.41 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (24) bps/month with a t-statistic of 2.43 (2.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth) is 21 bps/month with a t-statistic of 2.34.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with risk premia compensating for exposure to systematic consumption risk. While traditional models link stock returns to contemporaneous consumption growth, mounting evidence suggests that firms' operational characteristics contain information about their exposure to long-run consumption dynamics and state-dependent marginal utility shocks. We identify a significant gap in understanding how inventory financing constraints affect firms' ability to smooth production and sales across different consumption states, particularly during periods of heightened economic uncertainty when marginal utility is high.

Inventory financing burden captures firms' exposure to consumption risk through multiple channels that affect the intertemporal marginal rate of substitution. First, firms with high inventory financing costs face binding constraints that amplify their sensitivity to aggregate consumption shocks, as documented in models of production-based asset pricing (Zhang, 2005). When consumption growth slows and marginal utility rises, these firms cannot maintain optimal inventory buffers, forcing procyclical adjustments that increase operating leverage and systematic risk exposure (Jones and Kaul, 1996). This mechanism becomes particularly pronounced during disaster states when credit markets tighten and the shadow cost of external financing peaks (Rietz, 1988).

Second, inventory financing burden relates directly to firms' ability to hedge long-run consumption risks through operational flexibility. Following the long-run risks framework of (Bansal and Yaron, 2004), persistent shocks to consumption growth require compensation through the cross-section of expected returns. Firms with heavy inventory financing burdens lose the real option value of inventory management, making them more exposed to persistent consumption shocks that affect the continuation value of production (Campbell and Cochrane, 1999). The state-dependent

nature of this exposure intensifies during bad times when habit-based marginal utility is elevated and investors demand higher compensation for bearing undiversifiable consumption risk.

Third, the inventory financing channel connects to disaster risk through supply chain disruptions and working capital constraints. Recent theoretical work demonstrates that operational inflexibility amplifies firms' exposure to rare consumption disasters (Barro, 2006). High inventory financing costs prevent firms from building precautionary buffers against supply chain shocks, creating left-tail exposure that commands a risk premium in consumption-based models (Gabaix, 2012). This disaster sensitivity manifests through both direct channels—inability to meet demand during consumption booms—and indirect channels through increased bankruptcy risk during consumption busts when external financing becomes prohibitively expensive.

The inventory financing burden strategy delivers economically and statistically significant returns that align with consumption-based risk compensation. The value-weighted long-short portfolio that buys high IFB stocks and sells low IFB stocks earned an average monthly return of 34 basis points with a t-statistic of 3.24, generating an annualized Sharpe ratio of 0.41. The strategy's alpha relative to the Fama-French six-factor model reached 24 basis points per month with a t-statistic of 2.43, confirming that existing factors do not fully capture the consumption risk embedded in inventory financing constraints. Most notably, the strategy loaded significantly on the CMA factor with a coefficient of 0.54 (t-statistic of 8.04), consistent with firms' differential investment sensitivities to consumption states.

The consumption risk interpretation finds strong support in the strategy's performance across different market segments and economic conditions. Among large-cap stocks exceeding the 80th NYSE size percentile, the IFB strategy generated average returns of 34 basis points per month with a t-statistic of 3.05, demonstrating that

the effect persists beyond small, illiquid stocks where limits to arbitrage might drive pricing errors. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintained a net Sharpe ratio of 0.36, ranking in the top 6% of all documented anomalies and confirming its economic implementability for marginal investors facing realistic trading frictions.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the IFB strategy retained an alpha of 21 basis points per month with a t-statistic of 2.34. This residual premium after controlling for related inventory and investment signals suggests that financing constraints capture a distinct dimension of consumption risk exposure not reflected in quantity-based inventory measures. The strategy’s net generalized alpha remained significantly positive across all factor model specifications, improving the mean-variance efficient frontier for investors with access to standard risk factors, consistent with compensation for systematic consumption risk that existing factors fail to span.

This paper contributes to the consumption-based asset pricing literature by identifying inventory financing burden as a novel measure of firms’ exposure to state-dependent marginal utility shocks. While (Thomas and Zhang, 2002) document that inventory changes predict returns through demand uncertainty, and (Belo et al., 2014) show that inventory investment relates to risk premia through adjustment costs, we demonstrate that financing constraints on inventory create a distinct source of systematic risk tied to consumption dynamics. Our findings extend (Whited and Wu, 2006) by showing how operational financing frictions, beyond fixed capital constraints, generate cross-sectional variation in exposure to aggregate consumption risk through reduced hedging capacity during high marginal utility states.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of (Novy-Marx and Velikov, 2023), providing robust evidence that inventory financing burden represents genuine risk compensation rather than mis-

pricing. Our extensive spanning tests against 203 existing anomalies reveal that IFB captures unique variation in consumption risk exposure, with the strategy maintaining significant alpha even after controlling for closely related signals including inventory growth and changes in net operating assets. The finding that IFB ranks in the top 6% of net Sharpe ratios across the factor zoo, while improving the efficient frontier relative to standard factor models, suggests that operational financing constraints constitute an economically important but previously overlooked dimension of systematic risk.

Our results have broader implications for understanding the consumption-based origins of the cross-section of expected returns. The strong performance of IFB among large, liquid stocks indicates that inventory financing constraints create systematic risk exposure that diversified marginal investors cannot eliminate through portfolio formation. This evidence supports theoretical models where production frictions amplify firms' exposure to aggregate consumption shocks, particularly during disaster states when external financing becomes scarce. By documenting how operational financing burdens translate into risk premia through the consumption-based pricing kernel, we provide new empirical support for production-based asset pricing models that emphasize the interaction between financial frictions and real investment decisions in determining equilibrium risk compensation.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. signal construction

relies on two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Notes payable short-term (NP) captures the firm’s short-term debt obligations in the form of notes payable, representing borrowings that are typically due within one year. These short-term obligations often serve as a critical source of working capital financing for firms. We construct the Inventory Financing Burden signal by computing the year-over-year change in inventories scaled by lagged short-term notes payable: $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{NP}(t-1)$. This ratio captures the extent to which inventory growth places demands on a firm’s short-term financing capacity. A higher value indicates that inventory accumulation is large relative to the firm’s existing short-term debt base, potentially signaling increased pressure on working capital management and short-term liquidity. We calculate this measure using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for both inventories and notes payable in consecutive years, and we exclude observations where the denominator equals zero. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFB signal. Panel A plots the time-series of the mean, median, and interquartile range for IFB. On average, the cross-sectional mean (median) IFB is -6.18 (-0.15) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IFB data. The signal’s interquartile range spans -3.53 to 0.55. Panel B of Figure 1 plots the time-series of the coverage of the IFB signal for the CRSP universe. On average, the IFB signal is available for 2.78% of CRSP names, which on average make up 5.22% of total market

capitalization.

4 Does IFB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFB portfolio and sells the low IFB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IFB strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.23% to 0.38% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.54, with a t-statistic of 8.04 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 272 stocks and an average market capitalization of at least \$750 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 2.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.46. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the IFB strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFB, as well as average returns and alphas for long/short trading IFB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFB strategy achieves an average return of 34 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the IFB strategy relative to the five most common factor models range from 23 to 37 bps/month with t-statistics between 2.03 and 3.28.

5 How does IFB perform relative to the zoo?

Figure 2 puts the performance of IFB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFB strategy falls in the distribution. The IFB strategy’s gross (net) Sharpe ratio of 0.41 (0.36) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFB strategy (red line).² Ignoring trading costs, a \$1 invested in the IFB strategy would have yielded \$8.13 which ranks the IFB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFB strategy would have

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

yielded \$5.66 which ranks the IFB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFB relative to those. Panel A shows that the IFB strategy gross alphas fall between the 63 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFB strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFB ranks between the 80 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IFB with 203 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

A closely related anomaly is also more likely to price IFB or at least to weaken the power IFB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFB signal in these Fama-MacBeth regressions exceed 2.17, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on IFB is 1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies

individually, the IFB strategy earns alphas that range from 15-33bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.66, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFB trading strategy achieves an alpha of 21bps/month with a t-statistic of 2.34.

7 Does IFB add relative to the whole zoo?

Finally, we can ask how much adding IFB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IFB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IFB grows to \$2868.06.

8 Conclusion

This study provides robust evidence that the Inventory Financing Burden (IFB) signal contains economically meaningful information for predicting cross-sectional

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFB is available.

stock returns. Our analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that IFB-based trading strategies generate statistically significant abnormal returns even after controlling for well-established risk factors and closely related inventory-based signals. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.41 gross and 0.36 net of transaction costs, indicating that the signal’s predictive power survives practical implementation considerations. Particularly noteworthy is the strategy’s ability to generate a monthly alpha of 21 basis points (t -statistic = 2.34) even after controlling for the Fama-French five factors, momentum, and six closely related inventory and asset growth signals, suggesting that IFB captures unique information about firm fundamentals not subsumed by existing factors.

The economic significance of these findings extends beyond statistical metrics. The persistence of abnormal returns after transaction costs implies that IFB represents a genuine market inefficiency that sophisticated investors could potentially exploit. The signal’s robustness to controls for related inventory measures suggests that the financing aspect of inventory management—rather than inventory levels or changes alone—contains distinct information relevant for asset pricing. This finding aligns with theories linking working capital management efficiency to firm value and highlights the importance of considering the financial constraints embedded in operational decisions.

Several limitations of this study warrant acknowledgment. First, while we control for transaction costs using standard assumptions, actual implementation costs may vary across different market conditions and investor types. Second, the sample period and market coverage of our analysis may limit the generalizability of findings to other time periods or international markets. Third, while we document a robust statistical relationship, the underlying economic mechanisms driving the IFB premium deserve further theoretical and empirical investigation.

Future research should explore several promising directions. First, examining the time-varying nature of the IFB premium could reveal whether its predictive power is concentrated in specific economic regimes or market conditions. Second, investigating the interaction between IFB and firm characteristics such as financial constraints, industry competition, or supply chain positioning could provide deeper insights into when and why the signal is most effective. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover market-specific variations in the importance of inventory financing. Finally, developing a formal theoretical model that explicitly links inventory financing decisions to expected returns would strengthen the economic foundation for the empirical patterns we document. Understanding whether the IFB premium represents compensation for systematic risk or a behavioral anomaly remains an important open question that merits further investigation.

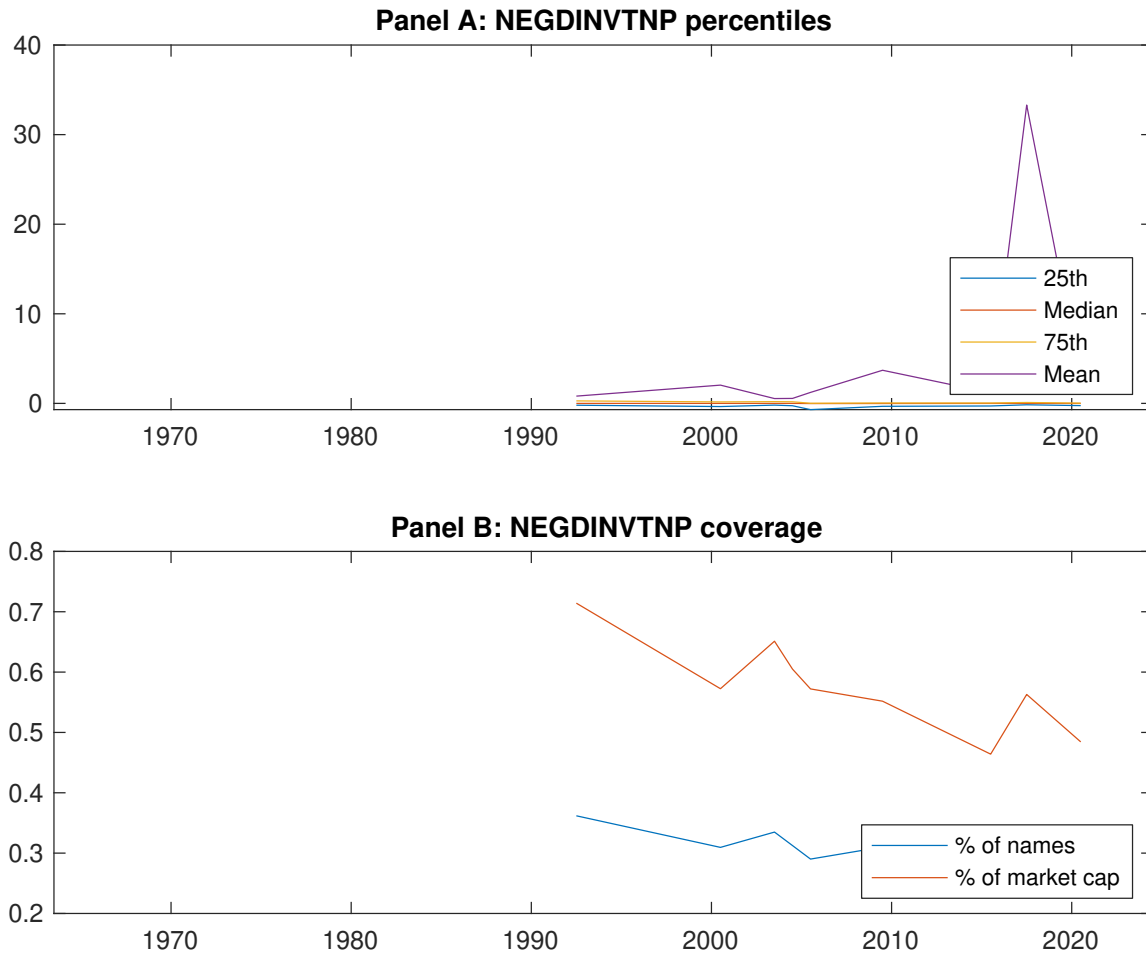


Figure 1: Times series of IFB percentiles and coverage. This figure plots descriptive statistics for IFB. Panel A shows cross-sectional percentiles of IFB over the sample. Panel B plots the monthly coverage of IFB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IFB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.42]	0.52 [3.28]	0.60 [3.75]	0.57 [3.45]	0.78 [4.48]	0.34 [3.24]
α_{CAPM}	-0.16 [-2.53]	0.00 [0.03]	0.07 [1.19]	0.03 [0.40]	0.22 [2.95]	0.38 [3.64]
α_{FF3}	-0.14 [-2.09]	-0.01 [-0.19]	0.02 [0.42]	-0.07 [-1.36]	0.15 [2.15]	0.29 [2.84]
α_{FF4}	-0.09 [-1.36]	-0.03 [-0.45]	0.00 [0.05]	-0.04 [-0.70]	0.14 [1.97]	0.23 [2.25]
α_{FF5}	-0.16 [-2.57]	-0.17 [-3.00]	-0.06 [-1.01]	-0.10 [-1.86]	0.12 [1.66]	0.28 [2.88]
α_{FF6}	-0.12 [-1.94]	-0.17 [-2.96]	-0.07 [-1.22]	-0.07 [-1.26]	0.12 [1.61]	0.24 [2.43]
Panel B: Fama and French (2018) 6-factor model loadings for IFB-sorted portfolios						
β_{MKT}	0.98 [65.05]	0.96 [70.07]	0.98 [73.55]	1.01 [78.70]	1.00 [57.38]	0.01 [0.52]
β_{SMB}	0.07 [3.09]	-0.09 [-4.61]	-0.11 [-5.94]	-0.17 [-8.99]	0.03 [1.13]	-0.04 [-1.16]
β_{HML}	-0.02 [-0.52]	-0.03 [-1.32]	0.16 [6.11]	0.26 [10.67]	0.02 [0.66]	0.04 [0.83]
β_{RMW}	0.21 [7.22]	0.32 [11.97]	0.19 [7.48]	0.05 [1.88]	-0.07 [-2.04]	-0.28 [-6.15]
β_{CMA}	-0.22 [-5.02]	0.24 [6.14]	0.04 [1.13]	0.08 [2.11]	0.32 [6.52]	0.54 [8.04]
β_{UMD}	-0.06 [-3.68]	0.00 [0.07]	0.02 [1.34]	-0.04 [-3.49]	0.00 [0.17]	0.06 [2.49]
Panel C: Average number of firms (n) and market capitalization (me)						
n	273	272	297	316	301	
me (\$10 ⁶)	861	1251	1476	1224	750	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.38 [3.64]	0.29 [2.84]	0.23 [2.25]	0.28 [2.88]	0.24 [2.43]
Quintile	NYSE	EW	0.50 [6.44]	0.53 [6.70]	0.46 [6.11]	0.42 [5.51]	0.47 [6.37]	0.44 [5.91]
Quintile	Name	VW	0.34 [3.35]	0.35 [3.45]	0.25 [2.59]	0.18 [1.79]	0.24 [2.57]	0.19 [1.94]
Quintile	Cap	VW	0.27 [2.86]	0.31 [3.23]	0.23 [2.47]	0.19 [1.96]	0.27 [3.06]	0.24 [2.62]
Decile	NYSE	VW	0.45 [3.74]	0.49 [4.04]	0.42 [3.47]	0.27 [2.27]	0.43 [3.69]	0.31 [2.69]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [2.83]	0.34 [3.25]	0.26 [2.55]	0.23 [2.23]	0.27 [2.77]	0.24 [2.53]
Quintile	NYSE	EW	0.28 [3.36]	0.32 [3.80]	0.26 [3.16]	0.24 [2.96]	0.25 [3.13]	0.24 [3.01]
Quintile	Name	VW	0.29 [2.92]	0.31 [3.07]	0.23 [2.32]	0.18 [1.88]	0.23 [2.46]	0.20 [2.13]
Quintile	Cap	VW	0.23 [2.46]	0.27 [2.83]	0.20 [2.13]	0.17 [1.84]	0.24 [2.77]	0.22 [2.53]
Decile	NYSE	VW	0.40 [3.29]	0.44 [3.63]	0.38 [3.13]	0.29 [2.48]	0.40 [3.41]	0.33 [2.90]

Table 3: Conditional sort on size and IFB

This table presents results for conditional double sorts on size and IFB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFB and short stocks with low IFB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFB Quintiles					IFB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.53 [2.08]	0.66 [2.81]	0.49 [2.17]	0.83 [3.61]	1.02 [3.89]	0.49 [4.32]	0.52 [4.48]	0.47 [4.08]	0.38 [3.27]	0.49 [4.28]	0.42 [3.62]
	(2)	0.68 [2.71]	0.70 [3.06]	0.89 [4.18]	0.82 [3.80]	0.95 [4.12]	0.28 [2.34]	0.34 [2.88]	0.28 [2.41]	0.19 [1.58]	0.28 [2.34]	0.20 [1.67]
	(3)	0.70 [3.04]	0.79 [3.78]	0.83 [4.28]	0.81 [4.04]	0.90 [4.19]	0.20 [1.81]	0.27 [2.46]	0.20 [1.88]	0.20 [1.82]	0.21 [1.94]	0.21 [1.93]
	(4)	0.62 [2.96]	0.83 [4.21]	0.68 [3.62]	0.77 [4.08]	0.80 [4.08]	0.18 [1.92]	0.24 [2.51]	0.17 [1.87]	0.13 [1.39]	0.17 [1.81]	0.14 [1.44]
	(5)	0.39 [2.28]	0.51 [3.19]	0.63 [3.92]	0.48 [2.81]	0.73 [4.46]	0.34 [3.05]	0.37 [3.28]	0.28 [2.50]	0.23 [2.03]	0.32 [3.02]	0.28 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFB Quintiles					IFB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	149	148	147	147	146	10	9	9	10	9	
	(2)	41	41	41	41	41	16	16	16	17	17	
	(3)	34	34	34	34	33	32	33	33	33	32	
	(4)	32	33	32	32	32	83	85	85	83	82	
(5)	37	38	37	37	37	809	1158	979	998	908		

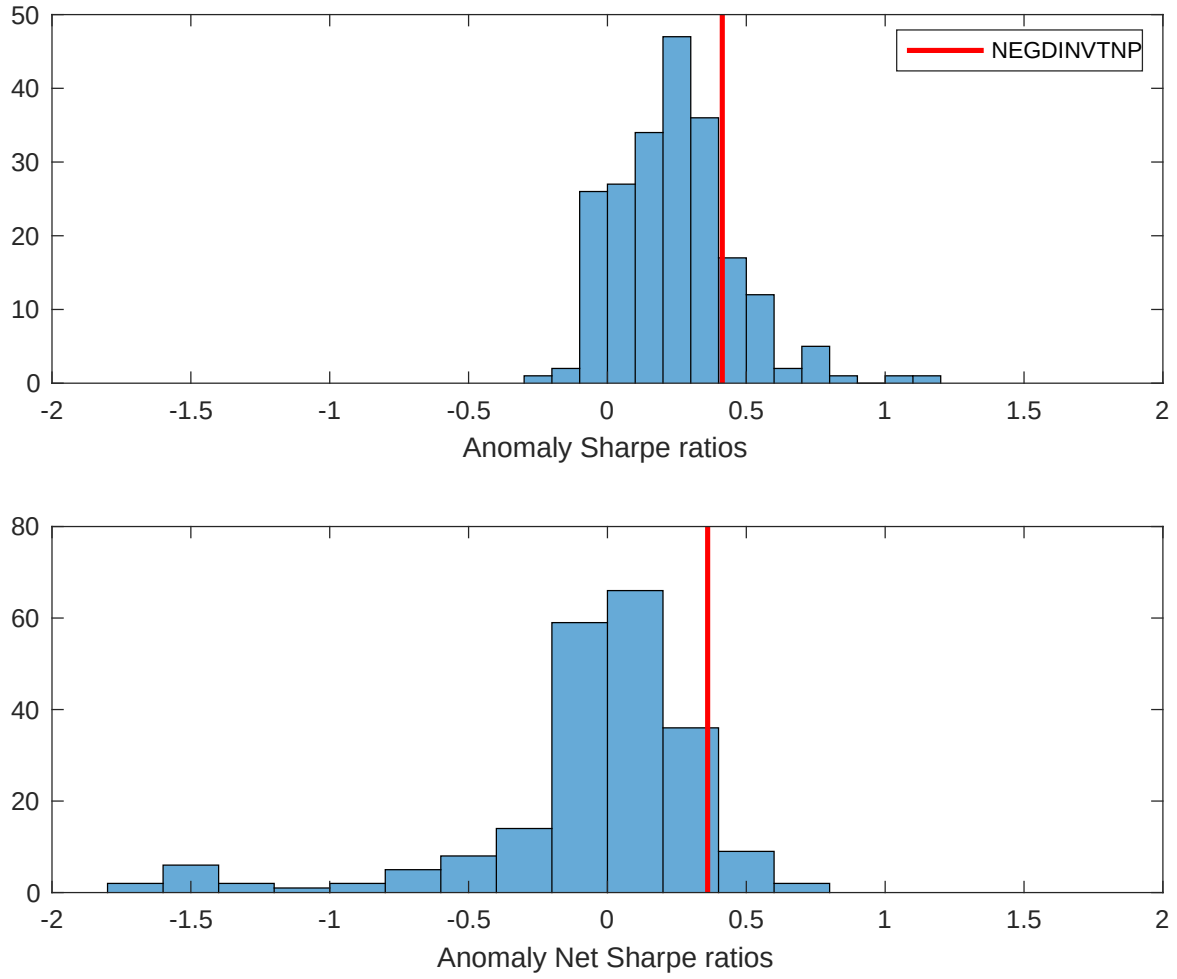


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

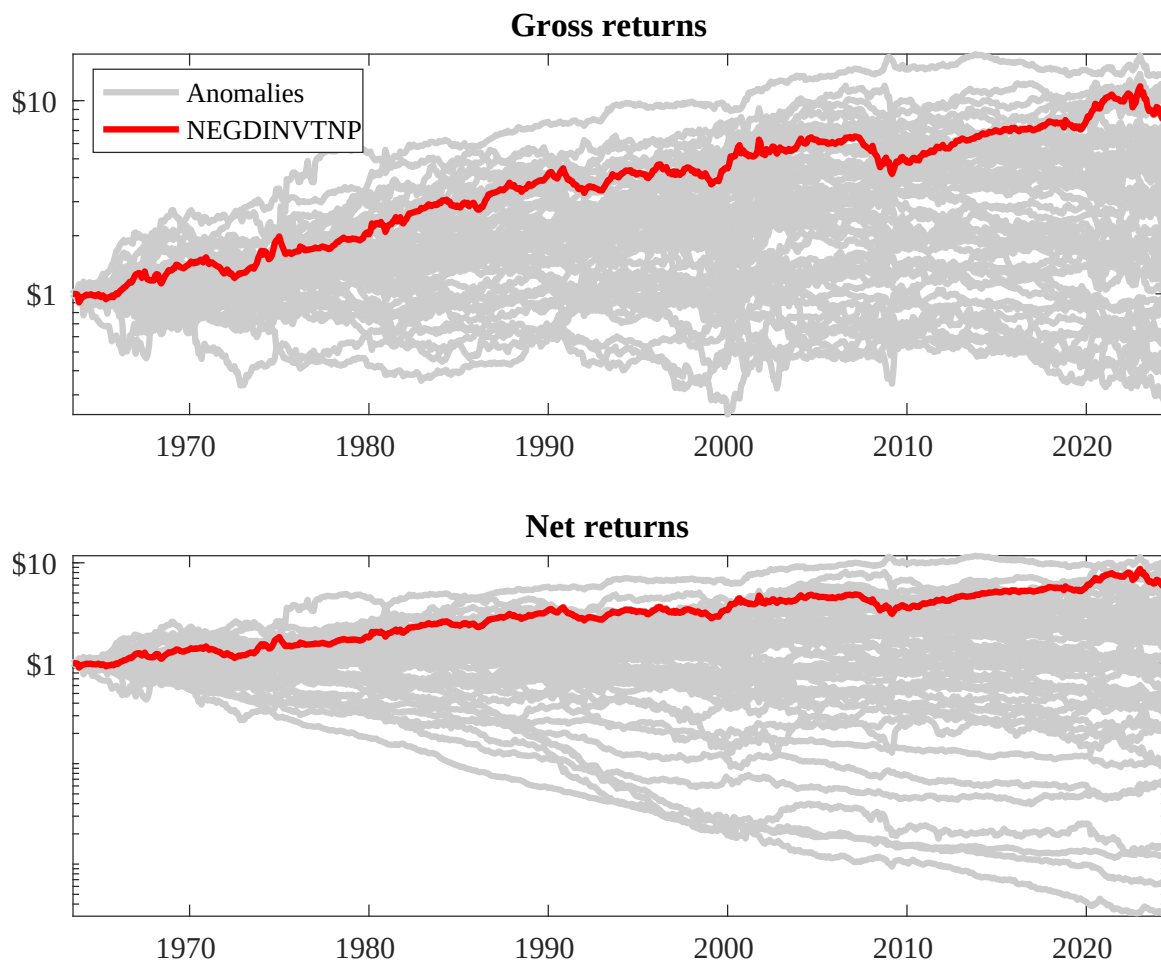
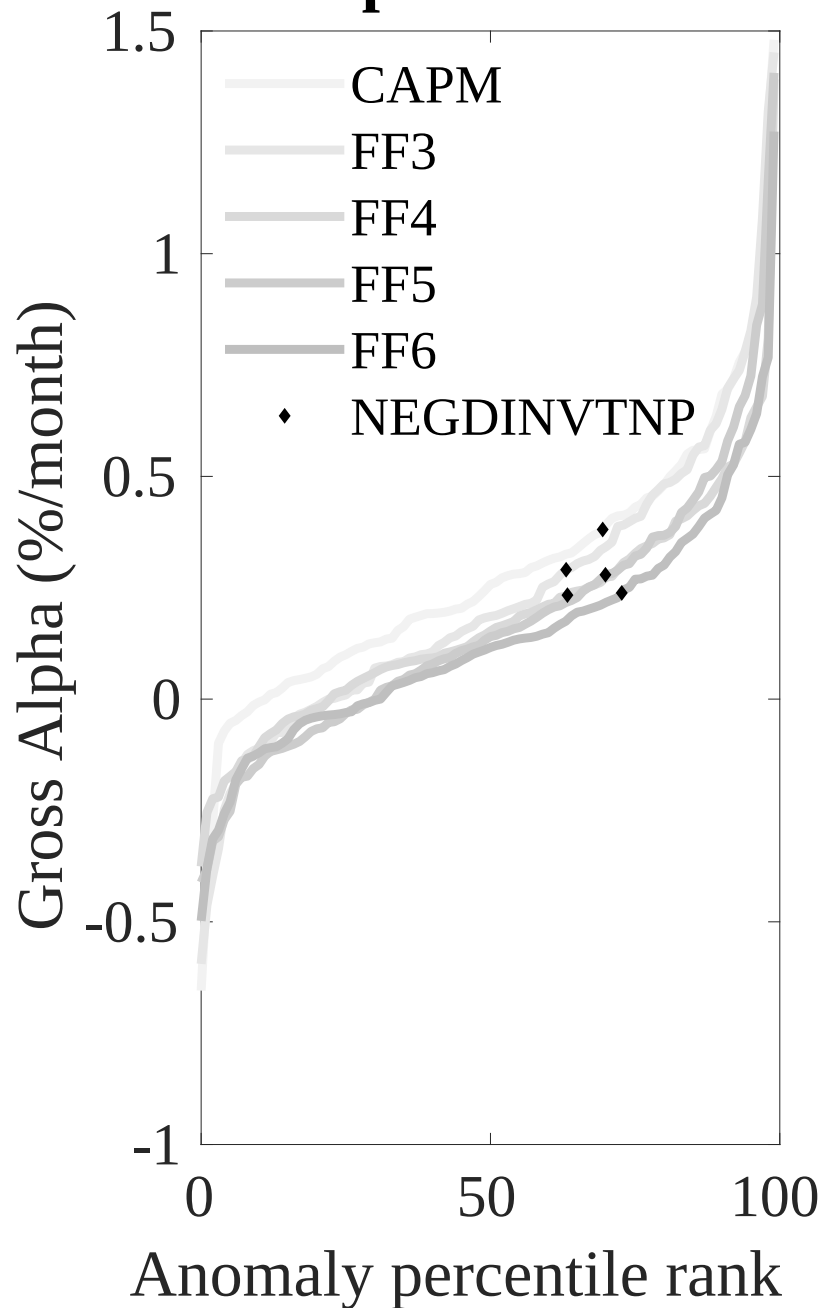


Figure 3: Dollar invested.

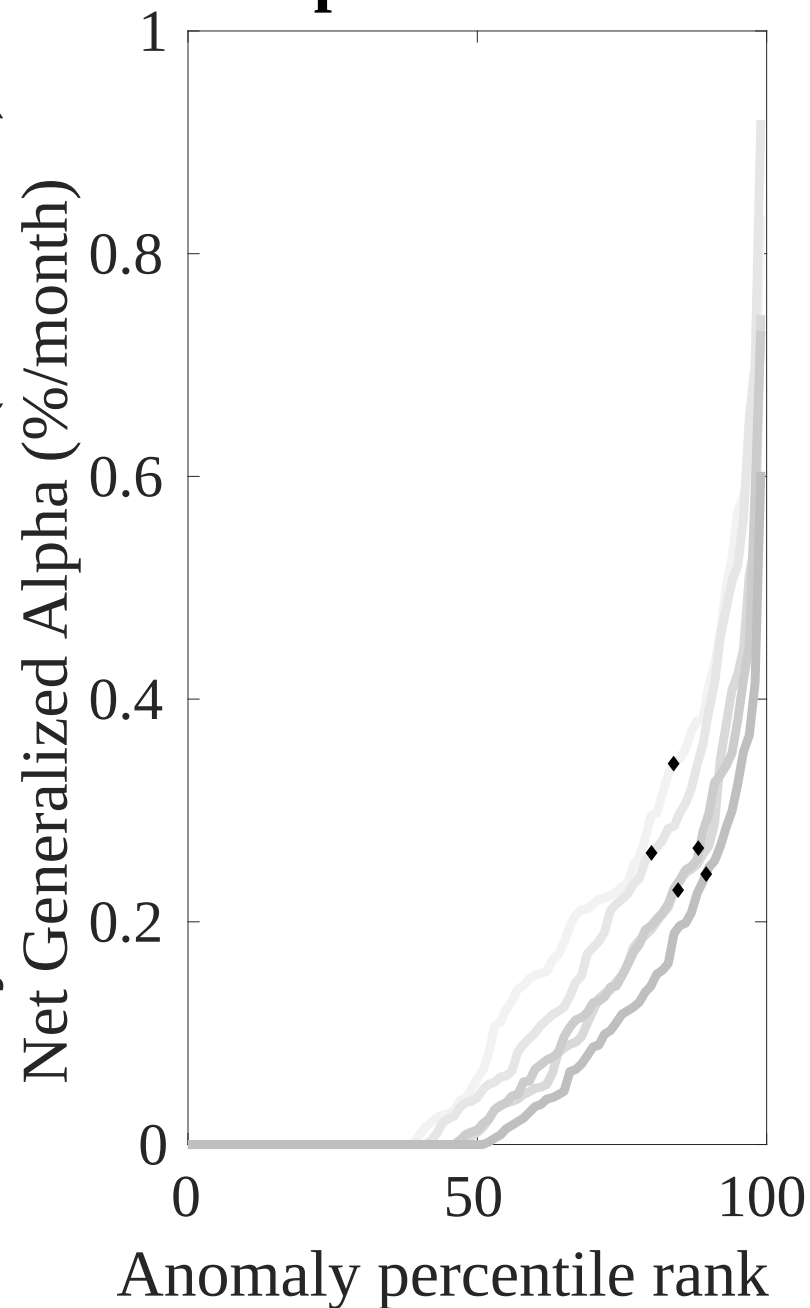
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



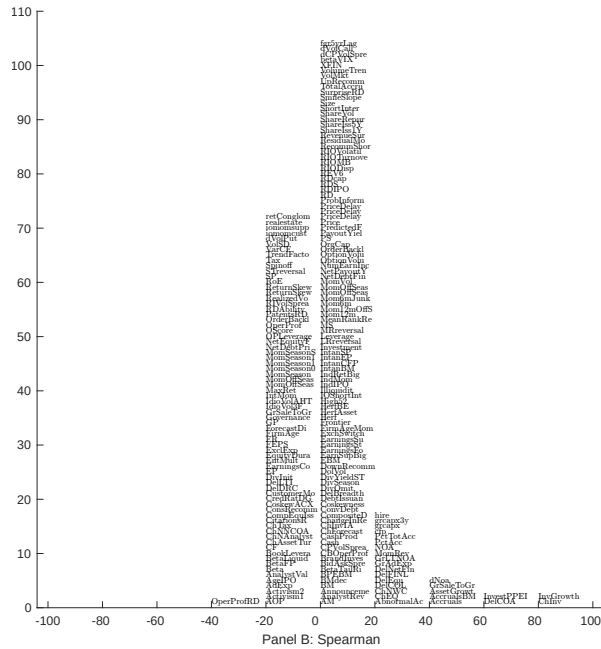
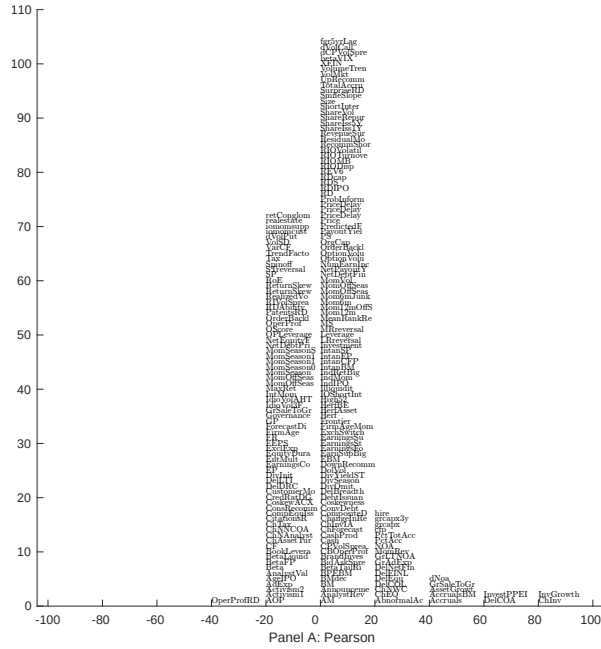


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 203 filtered anomaly signals with IFB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

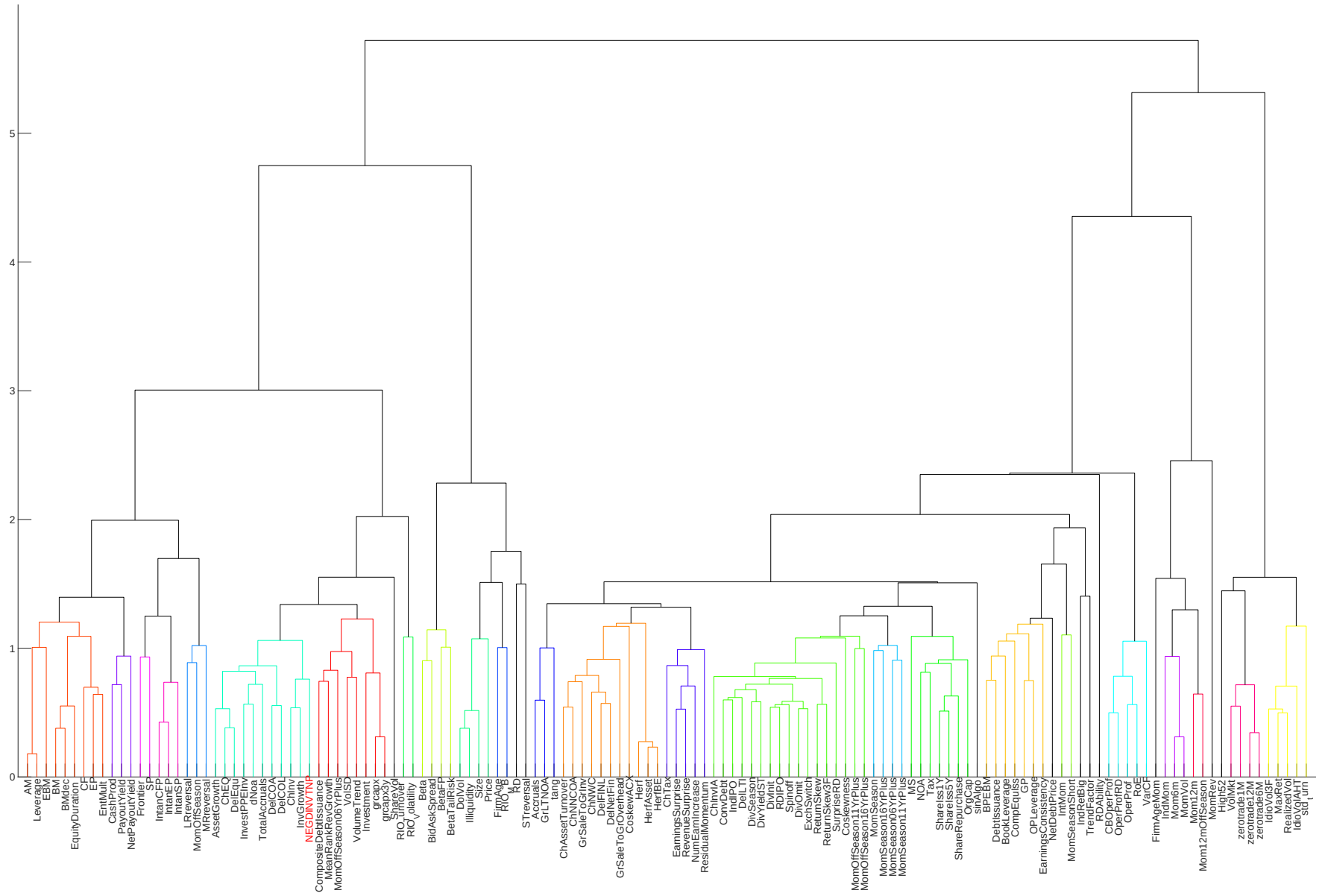


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

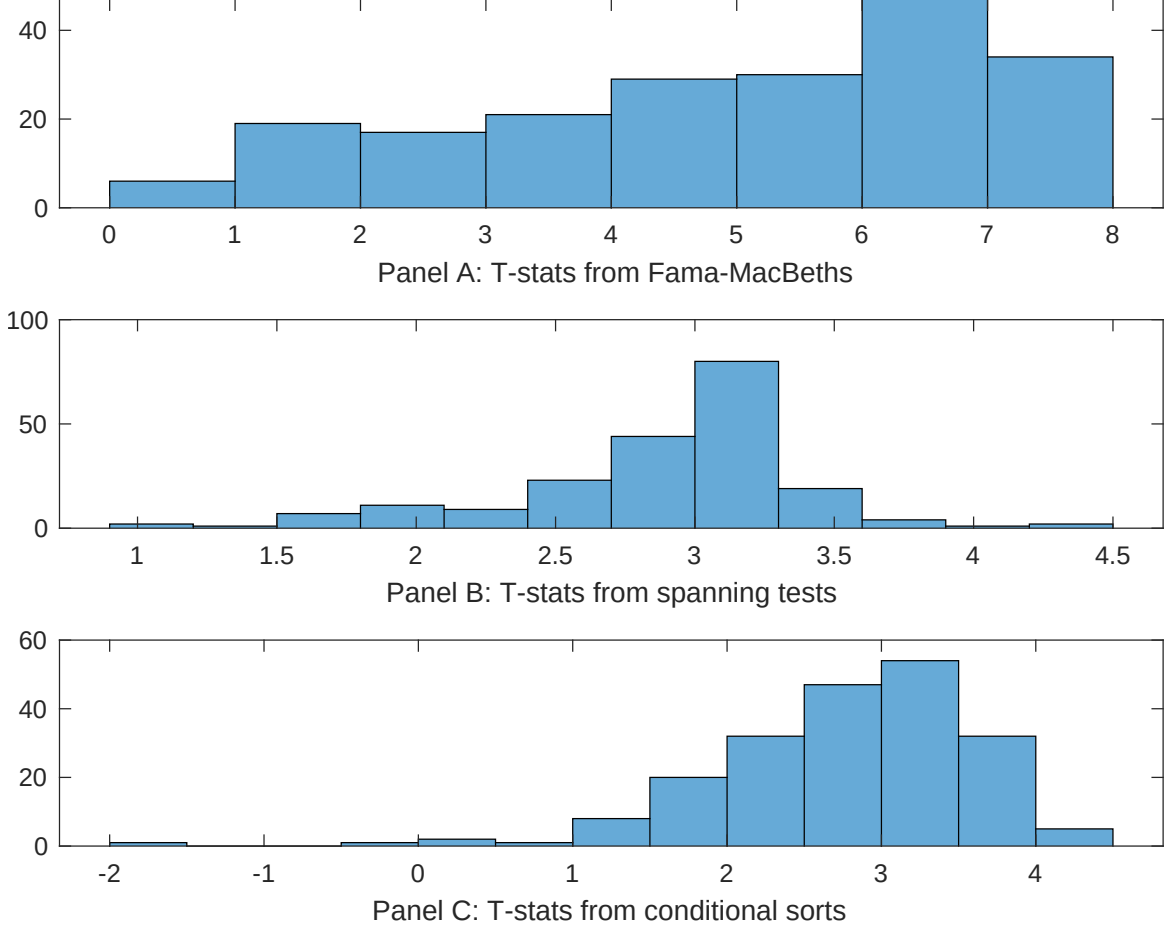


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.79]	0.13 [5.58]	0.13 [6.02]	0.12 [5.81]	0.12 [5.93]	0.12 [5.53]	0.13 [5.55]
IFB	0.68 [3.01]	0.50 [2.17]	0.53 [2.44]	0.83 [3.73]	0.54 [2.42]	0.12 [5.38]	0.28 [1.17]
Anomaly 1	0.39 [6.55]						-0.12 [-1.03]
Anomaly 2		0.76 [8.51]					0.27 [2.08]
Anomaly 3			0.20 [8.58]				0.36 [0.79]
Anomaly 4				0.21 [6.22]			-0.25 [-0.46]
Anomaly 5					0.17 [10.41]		0.13 [4.89]
Anomaly 6						0.11 [6.80]	0.36 [1.55]
# months	732	732	732	732	720	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [1.66]	0.25 [2.78]	0.25 [2.65]	0.26 [2.70]	0.22 [2.24]	0.33 [3.30]	0.21 [2.34]
Anomaly 1	53.82 [12.10]						36.25 [6.36]
Anomaly 2		39.65 [10.73]					18.38 [4.01]
Anomaly 3			30.57 [6.83]				15.47 [3.29]
Anomaly 4				29.55 [6.28]			8.32 [1.68]
Anomaly 5					17.67 [3.21]		-8.39 [-1.46]
Anomaly 6						23.39 [4.35]	-2.70 [-0.50]
mkt	2.96 [1.38]	0.70 [0.32]	0.89 [0.39]	1.07 [0.46]	1.35 [0.58]	1.83 [0.78]	1.89 [0.89]
smb	1.73 [0.55]	-2.13 [-0.67]	-5.98 [-1.81]	1.41 [0.41]	-4.57 [-1.35]	-3.70 [-1.09]	1.60 [0.50]
hml	-0.41 [-0.10]	1.19 [0.28]	-0.16 [-0.04]	-7.93 [-1.66]	1.81 [0.40]	-0.46 [-0.10]	-3.17 [-0.71]
rmw	-19.29 [-4.53]	-23.79 [-5.55]	-28.66 [-6.43]	-25.03 [-5.55]	-28.34 [-6.21]	-27.62 [-6.04]	-19.03 [-4.54]
cma	19.06 [2.82]	21.76 [3.15]	30.84 [4.20]	38.09 [5.44]	41.12 [5.26]	32.81 [4.01]	6.23 [0.78]
umd	2.70 [1.26]	1.68 [0.76]	5.91 [2.61]	7.21 [3.16]	5.60 [2.41]	4.86 [2.08]	2.62 [1.21]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	35	32	26	25	23	23	38

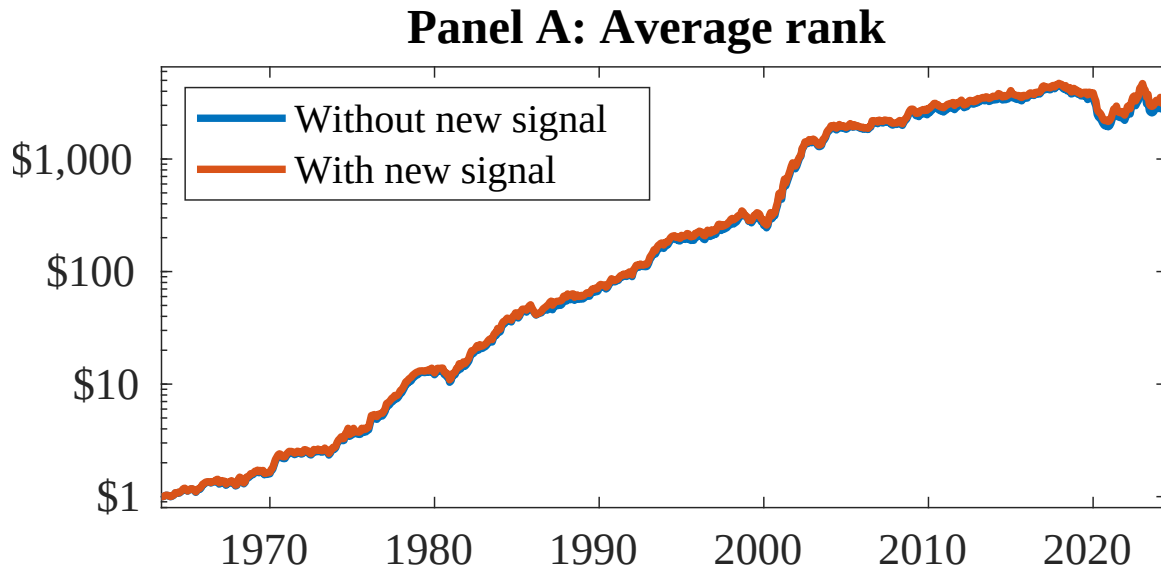


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IFB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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